

One Fold for Haar Oriented Two-Planes in Every Dimension $n \geq 6$: A Phase-Space Proof through a Fano Factor, an Integrating Factor and the Threshold $L_q = q(q - 1)/8$

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Abstract

Let $L_q(\kappa) = q! \sum_{j \geq 0} \kappa^{2j} / (q + 2j)!$, $q = n - 2$, be the normalizer of Haar oriented two-planes in $\mathbb{R}^n$, $m_q = L'_q / L_q$, and $H_q = m_q - \kappa m'_q$ the fold numerator of the stationary multiplier $\lambda_q = \kappa / (2m_q)$. The record ARR-2026-7H9FAPTBZA897AMJ proved that H_q has exactly one positive zero for every $q \geq 4$ through a one-change law for the coefficients of $L_q^2 H_q$. We give a short independent proof by a different method: in the coordinates $(X, u) = (\kappa m_q, 1/L_q)$ the inhomogeneous second-order equation of L_q makes κH_q a polynomial, the transversality of a crossing is a second polynomial T , and $Q = T - (q - \frac{1}{2})\kappa H_q$ obeys a first-order linear equation whose forcing, on the zero set of H_q , changes sign exactly where $L_q = q(q - 1)/8$. An integrating factor then excludes a zero before that point and a second zero after the first. Probabilistically, $\kappa H_q = 4(\mathbb{E}J - \text{Var}J)$ for the index law of the series. The same argument gives no fold for $q = 2, 3$. A now redundant interval-arithmetic certificate and numerics are included.

1. Introduction

Antecedents. For a Haar oriented two-plane in $\mathbb{R}^n$ with signed Plücker coordinate $W = x \wedge y$, the record ARR-2026-7H9FAPTBZA897AMJ (below 7H9F) reduced the unrestricted Shannon rate-distortion function under squared ambient loss to a one-dimensional radial problem whose normalizer is

$$L_q(\kappa) = q! \sum_{j \geq 0} \frac{\kappa^{2j}}{(q + 2j)!} = (n - 2) \int_0^1 \cosh(\kappa x) (1 - x)^{n-3} dx, \quad q = n - 2$$

(7H9F eqs. (11), (24)). Its complete phase diagram rests on one analytic fact: the stationary multiplier $\lambda_q(\kappa) = \kappa / (2m_q(\kappa))$, $m_q = L'_q / L_q$, has exactly one fold, i.e. the fold numerator $H_q = m_q - \kappa m'_q$ has exactly one positive zero. 7H9F proved this for every $q \geq 4$ ($n \geq 6$) as a consequence of its Theorem 8.2, the *one-change law*: writing $J_q = L_q^2 H_q = \sum_r c_{q,r} \kappa^{2r-1}$, the coefficients satisfy $c_{q,r} < 0$ for $2 \leq r < q$ and $c_{q,r} > 0$ for $r \geq q$ (with $c_{4,3} = 0$), proved by a likelihood-ratio ordering of a parity-truncated central binomial law and an exact tail estimate; the sign pattern of H_q then follows because $y^{-q} \sum_r c_{q,r} y^r$ ($y = \kappa^2$) is strictly increasing from $-\infty$ to $+\infty$. The cases $n = 3, 4, 5$ ($q = 1, 2, 3$) are 7H9F Lemma 7.1, where $H_q > 0$ on $(0, \infty)$ by explicit series of elementary functions. 7H9F Theorem 9.1 turns the one-zero statement into the complete high-dimensional phase: unique coexistence multiplier, one positive active radius above it, no later exchange or reentrance.

Two related records use a different tool. The published complex-projective one-fold paper ARR-2026-68BH5JDJCQ8XW8GT (below 68BH) proved one fold for ${}_1F_1(1; d; \kappa)$, $d \geq 3$, from the first-order inhomogeneous equation $\kappa M' = (\kappa - d + 1)M + (d - 1)$ and a transversality argument in the plane (M, κ) ; the unpublished manuscript [B3] extended this to ${}_1F_1(a; c; \kappa)$ by a Riccati reduction to the plane (m, κ) and proved “one fold iff $c > 2a$ ”. For the oriented-two-plane normalizer, which satisfies a homogeneous *third-order* equation, [B3] Section 8 obtained only an exact identity, $H'_q + (3m_q + (2q + 2)/\kappa)H_q = \Delta_q$, with an explicit $\Delta_q(m_q, \kappa)$, and numerics; it recorded the obstruction that at the fold the $O(q)$ terms of Δ_q cancel, so that no crude estimate of m_q decides the sign, and left a second proof of 7H9F Theorem 8.2 as its first open question. The record ARR-2026-61Y0FFA39M8KMBJ5 (below 61Y0) treats $\text{Gr}_{\mathbb{C}}(2, 4)$, whose normalizer equals L_4 , by a coefficient method and gives the numerical values at $q = 4$ used in Section 7.

Contribution. We prove the one-zero statement of 7H9F Theorem 8.2 for every $q \geq 4$ by a phase-space argument that uses no coefficient sign, no estimate of m_q or L_q , and nothing evaluated at the fold (Theorem 4.1). The ingredients are: (i) the inhomogeneous second-order equation (E2) of L_q , which descends from 68BH’s first-order equation for ${}_1F_1(1; q + 1; \kappa)$ because L_q is its even part; (ii) the coordinates $(X, u) = (\kappa m_q, 1/L_q)$, in which κH_q and the transversality of a crossing of $H_q = 0$ are polynomials h and T (identities (C1)–(C6), all machine-checked with symbolic q); (iii) the function $Q = T - (q - \frac{1}{2})h$, which satisfies a first-order linear inhomogeneous equation $\dot{Q} + cQ = g + hB$ with $g = X^2(8 - q(q - 1)u)$, $B \geq 0$ (Lemma 3.1); and (iv) the integrating factor κ^c , which shows that H_q cannot reach 0 from below while $L_q \leq q(q - 1)/8$ and cannot return to 0 from above once $L_q > q(q - 1)/8$. Probabilistically (Proposition 3.2), $\kappa H_q = 4(\mathbb{E}J - \text{Var}J)$ for the index law $P(J = j) \propto \kappa^{2j}/(q + 1)_{2j}$, so the fold is where the Fano factor of that law crosses 1, and the threshold $L_q = q(q - 1)/8$ is a condition on the boundary atom of a parity-conditioned Poisson law. The same argument with $c = 2$, $\frac{5}{2}$ recovers 7H9F Lemma 7.1 for $n = 4, 5$ (Corollary 5.1). We do **not** reprove the coefficient law $c_{q,r} \leq 0$ itself; what is reproved is its consequence, which is all that 7H9F Theorem 9.1 uses (Corollary 4.5). A rigorous interval-arithmetic certificate of the pre-threshold sign for $4 \leq q \leq 200$ (Section 6), written before Step 1 of the proof was found, is retained as a redundant check.

Status. Theorem 4.1 and Corollary 5.1 are proved; every identity of Section 3 is a short hand computation verified with symbolic q by the author’s and, independently, the reviewer’s scripts. An adversarial review re-derived every step and found no gap; its corrections (a bibliographic slip on modified Struve and Lommel functions, the transcription of one table entry, the wording of two remarks and of the claim of independence) are incorporated. Literature searches (Section 9) found no prior Lyapunov-type argument for the sign of $K' - \kappa K''$ of a hypergeometric normalizer; a null search does not certify priority.

2. Setting

Throughout $q \geq 1$ is an integer, $\kappa > 0$, $z = \kappa^2$, and

$$L_q(\kappa) = \sum_{j \geq 0} w_j \kappa^{2j}, \quad w_j = \frac{q!}{(q + 2j)!} = \frac{1}{(q + 1)_{2j}}, \quad K_q = \log L_q, \quad m_q = K'_q, \quad H_q = m_q - \kappa m'_q. \quad (2.1)$$

L_q is entire, even, with positive coefficients, $L_q(0) = 1$, strictly increasing to $+\infty$ on $[0, \infty)$; $L_q = {}_1F_2(1; \frac{q+1}{2}, \frac{q+2}{2}; \kappa^2/4)$, a modified Lommel function of the first kind with parameters $(\mu, \nu) = (q - \frac{3}{2}, \frac{1}{2})$ [Ga], and a modified Struve function exactly when a denominator parameter equals $\frac{3}{2}$, i.e. for $q = 1$ ($\sinh \kappa/\kappa$) and $q = 2$ [BP1, BP2]. The radial problem of 7H9F (its Sections 6–9) is $\max_{0 \leq b \leq 1} \{K_q(2\lambda b) -$

λb^2 }; a positive stationary radius satisfies $b = m_q(\kappa)$, $\lambda = \lambda_q(\kappa) := \kappa/(2m_q(\kappa))$, and (7H9F eqs. (22), (29))

$$\lambda'_q(\kappa) = \frac{H_q(\kappa)}{2m_q(\kappa)^2}, \quad F_q := 2K_q - \kappa m_q, \quad F'_q = H_q. \quad (2.2)$$

A *fold* of λ_q is a zero of H_q ; 7H9F's phase diagram uses only the sign pattern of H_q (Corollary 4.5). We use the closed form $L_q = q! \kappa^{-q} C_q(\kappa)$ with $C_q(\kappa) := \sum_{k \geq q, k \equiv q \pmod{2}} \kappa^k / k!$, so that $C'_q = C_{q-1}$ and, for $q \geq 2$, $C_{q-2} = C_q + \kappa^{q-2}/(q-2)!$; C_q is $\cosh \kappa$ or $\sinh \kappa$ minus a polynomial of degree at most $q-2$, hence $C_q = \frac{1}{2}e^\kappa (1 + O(\kappa^{q-2}e^{-\kappa}))$.

Phase-space variables. With $\tau = \log \kappa$ and $\dot{f} := df/d\tau = \kappa f'$,

$$u := \frac{1}{L_q} \in (0, 1), \quad X := \kappa m_q > 0, \quad h := \kappa H_q, \quad c := q - \frac{1}{2}. \quad (2.3)$$

Lemma 2.1 (elementary facts). (F1) $H_q = \gamma_q \kappa^3 + O(\kappa^5)$ with $\gamma_q = -4(q^2 - q - 8)/((q+1)^2(q+2)^2(q+3)(q+4))$; so $H_q < 0$ near 0^+ for $q \geq 4$ and $H_q > 0$ near 0^+ for $q \leq 3$. (F2) $H_q \rightarrow 1$ as $\kappa \rightarrow \infty$. (F3) $X > 0$ for $\kappa > 0$; u is a strictly decreasing bijection of $(0, \infty)$ onto $(0, 1)$. Hence for $q \geq 4$ there is a unique $\kappa_L > 0$ with

$$u(\kappa_L) = u_* := \frac{8}{q(q-1)} < 1, \quad \text{i.e.} \quad L_q(\kappa_L) = \frac{q(q-1)}{8}, \quad (2.4)$$

and $u > u_*$ on $(0, \kappa_L)$, $u < u_*$ on (κ_L, ∞) . (F4) $X \rightarrow 0$, $u \rightarrow 1$, $h \rightarrow 0$ and $T \rightarrow 0$ (with T from (3.4)) as $\kappa \rightarrow 0^+$.

Proof. (F1) is series substitution (`sym_reduction.py`, `rev_sym.py`); the coefficients of $\kappa^0, \kappa^1, \kappa^2$ vanish. (F2): $m_q = C_{q-1}/C_q - q/\kappa$; put $\rho = C_{q-1}/C_q = 1 + O(\kappa^{q-2}e^{-\kappa})$. Then $\rho' = C_{q-2}/C_q - \rho^2 = 1 - \rho^2 + \kappa^{q-2}/((q-2)!C_q)$ and

$$\kappa m'_q = \kappa(1 - \rho^2) + \frac{\kappa^{q-1}}{(q-2)!C_q} + \frac{q}{\kappa} \rightarrow 0,$$

so $H_q = m_q - \kappa m'_q \rightarrow 1$ (for $q = 1$ use $L_1 = \sinh \kappa / \kappa$ directly). (F3): L'_q has positive coefficients; L_q increases strictly from 1 to ∞ ; $q(q-1)/8 > 1$ iff $q \geq 4$. (F4) is immediate from $m_q = O(\kappa)$ and $u \rightarrow 1$. $\square$

3. The reduction

Proposition 3.1 (the inhomogeneous second-order equation). For every $q \geq 1$,

$$\kappa^2 L''_q + 2q\kappa L'_q - (\kappa^2 - q(q-1))L_q - q(q-1) = 0. \quad (E2)$$

Proof. The coefficient of κ^{2j} , $j \geq 1$, on the left is $[2j(2j-1) + 4qj + q(q-1)]w_j - w_{j-1} = (q+2j)(q+2j-1)w_j - w_{j-1} = 0$, and the constant term is $q(q-1)w_0 - q(q-1) = 0$. $\square$

(E2) is [B3] Proposition 8.1(ii); it is the even part of 68BH's first-order equation $\kappa E' = (\kappa - q)E + q$ for $E = {}_1F_1(1; q+1; \kappa)$ after elimination of the odd part, and its κ -derivative is the homogeneous third-order equation (8.1) of [B3]. Only (E2) is used below.

Proposition 3.2 (identities). Along $\kappa \mapsto (X(\kappa), u(\kappa))$, for every $q \geq 1$:

$$h = X^2 + (2q + 1)X + q(q - 1)(1 - u) - \kappa^2, \quad (C1)$$

$$\dot{X} = 2X - h, \quad \dot{u} = -Xu, \quad (C2)$$

$$\kappa^2 \Delta_q = T + (X + 2)h, \quad T := 2X^2 + q(q - 1)uX - 2q(q - 1)(1 - u), \quad (C3)$$

$$\dot{T} = X^2(8 - q(q - 1)u) - h(4X + q(q - 1)u), \quad (C4)$$

$$\dot{h} = T - (2X + 2q - 1)h, \quad (C5)$$

$$H'_q + \left(3m_q + \frac{2q + 2}{\kappa}\right)H_q = \Delta_q, \quad \Delta_q := -\kappa m_q(1 - m_q^2) + (2q + 5)m_q^2 - 2 + \frac{(q + 1)(q + 2)m_q}{\kappa}. \quad (C6)$$

Moreover $\dot{h} = \kappa(H_q + \kappa H'_q)$ and $\ddot{h} = \kappa^3 H''_q + 3\kappa^2 H'_q + \kappa H_q$.

Proof. Divide (E2) by L_q and use $L''_q/L_q = m'_q + m_q^2$: $\kappa^2 m'_q = \kappa^2 - X^2 - 2qX - q(q - 1)(1 - u)$. Then $h = X - \kappa^2 m'_q$ is (C1); $\dot{X} = X + \kappa^2 m'_q = 2X - h$ after eliminating κ^2 through (C1); $\dot{u} = -\kappa L'_q/L_q^2 = -Xu$. Differentiating (C1) with (C2) and eliminating κ^2 again gives (C5); differentiating T gives (C4). (C6) is [B3] eq. (8.2), and (C3) is (C6) rewritten in (X, u, h) ; both need the derivative of (E2). The last two formulas are $h = \kappa H_q$ differentiated. All identities are verified with symbolic q in `sym_reduction.py` and, independently, `rev_sym.py`; the residual of each is 0. $\square$

Reading of (C1) and (C3). By (C1), $h = (X - X_+)(X + X_+ + 2q + 1)$ where $X_+(\kappa)$ is the positive root of $X^2 + (2q + 1)X + q(q - 1)(1 - u) - \kappa^2 = 0$ (the constant term is negative for every $\kappa > 0$: comparing $L_q - 1$ with $\kappa^2 L_q / ((q + 1)(q + 2))$ termwise, $w_j \leq w_{j-1} / ((q + 1)(q + 2))$, gives $1 - u \leq \kappa^2 / ((q + 1)(q + 2))$, so the constant term is at most $-2(2q + 1)\kappa^2 / ((q + 1)(q + 2))$): the zeros of H_q are the crossings of the trajectory $X(\kappa)$ with the explicit curve $X_+(\kappa)$. At a zero of H_q , (C5) and the formula for $\dot{h}$ give $\kappa^2 H'_q = \dot{h} = T$: **T is the transversality of the crossing**, and by (C3) it equals $\kappa^2 \Delta_q$ there, matching [B3]. The dynamics (C2) is a closed system in (X, u, τ) ; κ^2 enters only through (C1).

Lemma 3.3 (the first-order equation for Q). Let $Q := T - c h$ with a constant c . Then

$$\dot{Q} + cQ = g + hB, \quad g := X^2(8 - q(q - 1)u), \quad B := (2c - 4)X + c(2q - 1 - c) - q(q - 1)u, \quad (3.1)$$

equivalently $\frac{d}{d\tau}(e^{c\tau}Q) = e^{c\tau}(g + hB)$ with $e^{c\tau} = \kappa^c$. For $c = q - \frac{1}{2}$ and $q \geq 3$, $B = (2q - 5)X + \frac{1}{4} + q(q - 1)(1 - u) > \frac{1}{4}$ for all $\kappa > 0$; for $(q, c) = (2, 2)$, $B = 2(1 - u) \geq 0$.

Proof. From (C4), (C5) and $-cT = -cQ - c^2 h$: $\dot{Q} = g - h(4X + q(q - 1)u) - cQ - c^2 h + c(2X + 2q - 1)h$, which is (3.1). For $c = q - \frac{1}{2}$: $2c - 4 = 2q - 5$ and $c(2q - 1 - c) = (q - \frac{1}{2})^2 = q(q - 1) + \frac{1}{4}$. $\square$

The forcing $g + hB$ is the whole mechanism: on the zero set $\{h = 0\}$ it reduces to g , whose sign is that of $u_* - u$, i.e. of $\kappa - \kappa_L$; off the zero set the term hB has the sign of h . The constant 8 is forced: for any $\tilde{Q}$ agreeing with T on $\{h = 0\}$, $\dot{\tilde{Q}}|_{h=0} = g + (\partial \tilde{Q} / \partial h) T$, so the forcing on the zero set is g up to a multiple of the transversality itself (a heuristic, not used).

Proposition 3.4 (probabilistic reading). Let J have the law $P(J = j) = w_j z^j / L_q$, $j \geq 0$ (an exponential family in $\theta = \log z$), and let $K \sim \text{Poisson}(\kappa)$ be conditioned on $A_q = \{K \geq q, K \equiv q \pmod{2}\}$, so that $K = q + 2J$ in law. Then

$$\mathbb{E}J = \frac{X}{2}, \quad \text{Var} J = \frac{d \mathbb{E}J}{d\theta} = \frac{\kappa}{4}(m_q + \kappa m'_q), \quad \kappa H_q = 4(\mathbb{E}J - \text{Var} J), \quad (3.2)$$

$$u = P(K = q \mid A_q), \quad X = \mathbb{E}[K - q \mid A_q], \quad \kappa H_q = 2 \mathbb{E}[K - q \mid A_q] - \text{Var}(K \mid A_q), \quad (3.3)$$

$$\mathbb{E}[K(K - 1) \mid A_q] = \kappa^2 + q(q - 1) P(K = q \mid A_q). \quad (3.4)$$

Hence $H_q < 0$ iff the Fano factor $\text{Var } J / \mathbb{E} J$ exceeds 1, the fold is where the conditional variance of K equals twice its conditional excess mean, and the threshold (2.4) is $P(K = q \mid A_q) = 8/(q(q - 1))$.

Proof. $\mathbb{E} J = z L'_q(z) / L_q$ in the variable z equals $\kappa L'_q(\kappa) / (2L_q) = X/2$; $d/d\theta = \frac{\kappa}{2} d/d\kappa$ gives the variance; then $4(\mathbb{E} J - \text{Var } J) = 2X - \kappa m_q - \kappa^2 m'_q = \kappa H_q$. $P(A_q) = e^{-\kappa} \kappa^q L_q / q!$, so $P(K = q \mid A_q) = 1/L_q$. For (3.4), $\sum_{k \in A_q} k(k - 1) \kappa^k / k! = \kappa^2 \sum_{k \in A_q - 2} \kappa^k / k!$ and $A_q - 2 = A_q \cup \{q - 2\}$, while $\kappa^2 P(K = q - 2) / P(K = q) = q(q - 1)$. $\square$

7H9F's parity-truncated binomial law concerns the coefficients $c_{q,r}$; the law of Proposition 3.4 is attached to the function itself. We are not aware of a theorem on a single crossing of the Fano factor of a power-series distribution with log-concave weights; the present proof gives one for the weights $1/(q + 1)_{2j}$.

4. Main theorem and proof

Theorem 4.1. Let $q \geq 4$ ($n = q + 2 \geq 6$) and let κ_L be defined by (2.4). Then H_q has exactly one positive zero κ_f ; $H_q < 0$ on $(0, \kappa_f)$, $H_q > 0$ on (κ_f, ∞) , $H'_q(\kappa_f) > 0$, and $\kappa_f > \kappa_L$. Consequently $\lambda_q = \kappa / (2m_q)$ is strictly decreasing on $(0, \kappa_f]$ and strictly increasing on $[\kappa_f, \infty)$: exactly one fold.

Throughout this section $c = q - \frac{1}{2}$, $Q = T - ch$, g and B are as in Lemma 3.3, and $B > \frac{1}{4}$.

Lemma 4.2 (Step 1: no zero before the threshold). $h < 0$ on $(0, \kappa_L]$.

Proof. Suppose not and let $\kappa_* := \inf\{\kappa \in (0, \kappa_L] : h(\kappa) \geq 0\}$. The set is closed in $(0, \kappa_L]$, so the infimum is attained and $h(\kappa_*) \geq 0$; by (F1) $\kappa_* > 0$ and $h < 0$ on $(0, \kappa_*)$, so by continuity $h(\kappa_*) = 0$. On $(0, \kappa_*]$ we have $h \leq 0$, hence $hB \leq 0$, and $u \geq u(\kappa_L) = u_*$ by (F3), hence $g = X^2(8 - q(q - 1)u) \leq 0$; by (3.1),

$$\frac{d}{d\tau}(e^{c\tau}Q) = e^{c\tau}(g + hB) \leq 0 \quad \text{on } (0, \kappa_*], \quad < 0 \text{ on } (0, \kappa_*)$$

(strictly because $X > 0$ and $u > u_*$ for $\kappa < \kappa_L$). Since $e^{c\tau}Q = \kappa^c Q \rightarrow 0$ as $\tau \rightarrow -\infty$ by (F4) and $c > 0$, integration gives $e^{c\tau_*}Q(\kappa_*) = \int_{-\infty}^{\tau_*} \frac{d}{ds}(e^{cs}Q) ds < 0$ (the improper integral of a sign-definite integrand converges because the left side is finite). Hence $T(\kappa_*) = Q(\kappa_*) + c h(\kappa_*) = Q(\kappa_*) < 0$. But $h < 0$ on $(0, \kappa_*)$ and $h(\kappa_*) = 0$ force $\dot{h}(\tau_*) \geq 0$, while (C5) gives $\dot{h}(\tau_*) = T(\kappa_*) - 0 < 0$. Contradiction. $\square$

Lemma 4.3 (Step 2: no second zero). Let κ_f be the first positive zero of H_q (it exists by (F1), (F2), and $\kappa_f > \kappa_L$ by Lemma 4.2). Then $H'_q(\kappa_f) > 0$, and H_q has no zero in (κ_f, ∞) .

Proof. Since $\kappa_f > \kappa_L$, (F3) gives $u < u_*$ on $[\kappa_f, \infty)$, so $g > 0$ there. $H_q < 0$ on $(0, \kappa_f)$ and $H_q(\kappa_f) = 0$ give $H'_q(\kappa_f) \geq 0$, i.e. $T(\kappa_f) = \kappa_f^2 H'_q(\kappa_f) \geq 0$ by the reading of (C5). A double zero is impossible: if $H_q(\kappa_f) = H'_q(\kappa_f) = 0$ then $h = T = 0$ there, and differentiating (C5) once more, $\ddot{h} = \dot{T} - (2X + 2q - 1)\dot{h} - h \frac{d}{d\tau}(2X + 2q - 1) = \dot{T} = g > 0$ at κ_f by (C4); but $\ddot{h} = \kappa^3 H''_q$ at such a point, so H_q would have a strict local minimum with value 0, contradicting $H_q < 0$ before κ_f . Hence $T(\kappa_f) > 0$ and $H_q > 0$ immediately after κ_f . Suppose H_q has a zero in (κ_f, ∞) and let κ_1 be the first one: the zeros of the nonconstant real-analytic function H_q are isolated and cannot accumulate at κ_f . On $[\kappa_f, \kappa_1]$ we have $h \geq 0$, hence $hB \geq 0$, and $g > 0$, so $\frac{d}{d\tau}(e^{c\tau}Q) \geq e^{c\tau}g > 0$ there and

$$Q(\kappa_1) > Q(\kappa_f) e^{-c(\tau_1 - \tau_f)} = T(\kappa_f) e^{-c(\tau_1 - \tau_f)} > 0.$$

But $h(\kappa_1) = 0$, so $Q(\kappa_1) = T(\kappa_1) = \kappa_1^2 H'_q(\kappa_1) \leq 0$ because H_q reaches 0 from above. Contradiction. $\square$

Proof of Theorem 4.1. Existence of κ_f and $\kappa_f > \kappa_L$: (F1), (F2), Lemma 4.2. Uniqueness, the signs, and $H'_q(\kappa_f) > 0$: Lemma 4.3. The statement on λ_q is (2.2) with $m_q > 0$. $\square$

Remarks 4.4. (i) *What is used.* Only (E2), the positivity of the coefficients of L_q , the sign of γ_q in (F1), and $H_q \rightarrow 1$. Nothing is estimated and nothing is evaluated at the fold; the obstruction recorded in [B3] Section 8 (cancellation of the $O(q)$ terms of Δ_q at the fold) does not arise because the sign of T is never needed on an interval, only at κ_* and κ_1 , where it comes from the integrating factor, and at κ_f , where it comes from $H'_q(\kappa_f) \geq 0$. The two halves of the argument are mirror images: before κ_L the forcing pushes $\kappa^c Q$ down, so H_q cannot come up to 0; after $\kappa_f > \kappa_L$ it pushes $\kappa^c Q$ up, so H_q cannot come down to 0. (ii) *Byproducts.* $T = Q + ch < 0$ on $(0, \kappa_L]$ (Step 1 gives $Q < 0$ there) and $T \geq Q > 0$ on $[\kappa_f, \infty)$; by (C3) the same signs hold for Δ_q along the trajectory. Hence *every* sign change of T or of Δ_q lies in (κ_L, κ_f) . That there is exactly one (numerically true, [B3] Section 8) is neither proved nor needed. (iii) *Relation to 7H9F Theorem 8.2.* Theorem 4.1 is the one-zero conclusion of that theorem, obtained without its coefficient law; it does not prove the law $c_{q,r} < 0$ ($2 \leq r < q$), $c_{q,r} > 0$ ($r \geq q$), which remains 7H9F's. (iv) *Recipe for other normalizers.* Write the fold numerator as $\kappa H = P(X, u, \kappa)$ with $(X, u) = (\kappa m, 1/L)$ using the lowest-order inhomogeneous equation available; compute the transversality polynomial $T = \dot{h}|_{h=0}$ and its forcing $\dot{T}|_{h=0}$; look for c with $B \geq 0$ and a forcing whose sign is governed by one monotone state variable. Whether this succeeds for other third-order normalizers is open (Section 8).

Corollary 4.5 (the chain to 7H9F Theorem 9.1). For every $n \geq 6$, the conclusions of 7H9F Theorem 9.1 (exactly one fold of λ_q ; a unique positive zero $\kappa_c > \kappa_f$ of F_q ; $0 < \lambda_c < n(n-1)/4$; below λ_c the zero radius is the unique global radius, at λ_c exactly 0 and $b_c = m_q(\kappa_c)$ are active, above it one positive radius is uniquely active, and there is no later exchange or reentrance; the two-piece rate-distortion function (30) of 7H9F) hold with Theorem 4.1 in place of 7H9F Theorem 8.2.

Proof. 7H9F's proof of Theorem 9.1 (its Section 9) uses from Theorem 8.2 only that H_q has a unique positive zero with $H_q < 0$ before and $H_q > 0$ after, through $\lambda'_q = H_q/(2m_q^2)$ and $F'_q = H_q$. Its other inputs — $F_q(0) = 0$, $F_q(\kappa) = \kappa - 2q \log \kappa + O(1) \rightarrow \infty$ (7H9F Proposition 11.1), the identities $\phi_\lambda(b) = F_q(\kappa)/2$ and $\phi''_\lambda(b) = 2\lambda(\kappa m'_q/m_q - 1)$ at a stationary radius, the inactivity of $b = 1$, and the positivity of the fourth cumulant for $n \geq 6$ — are independent of Theorem 8.2. Theorem 4.1 supplies exactly the sign pattern. $\square$

5. The corollary for $q = 2, 3$

Corollary 5.1 (7H9F Lemma 7.1 for $n = 4, 5$). For $q \in \{2, 3\}$, $H_q > 0$ on $(0, \infty)$; λ_q is strictly increasing and has no fold.

Proof. Take $c = 2$ for $q = 2$ and $c = \frac{5}{2}$ for $q = 3$; by Lemma 3.3, $B = 2(1-u) \geq 0$ and $B = X + \frac{1}{4} + 6(1-u) > 0$ respectively. Since $q(q-1) \leq 6 < 8$, $g = X^2(8 - q(q-1)u) > 0$ for every $\kappa > 0$ (there is no threshold), and $H_q > 0$ near 0^+ by (F1) ($q^2 - q - 8 = -6, -2$). If H_q had a zero, let κ_1 be the first; on $(0, \kappa_1]$, $h \geq 0$, so $\frac{d}{d\tau}(e^{c\tau}Q) \geq e^{c\tau}g > 0$, and with $e^{c\tau}Q \rightarrow 0$ at $-\infty$ we get $Q(\kappa_1) > 0$, i.e. $T(\kappa_1) = Q(\kappa_1) > 0$; but H_q reaches 0 from above, so $T(\kappa_1) = \kappa_1^2 H'_q(\kappa_1) \leq 0$. Contradiction. $\square$

For $q = 1$ ($n = 3$) the bracket B cannot be made nonnegative because $q(q-1) = 0$ kills the constant term; that case is 7H9F's explicit series for J_3 or the Bessel-family theorem of [B3] ($L_1 = {}_0F_1(\frac{3}{2}; \kappa^2/4)$).

6. The interval-arithmetic certificate (redundant)

Before Lemma 4.2 was found, the conclusion $H_q < 0$ on $(0, \kappa_L]$ was certified for individual q by rigorous interval arithmetic (`certify_neg.py`, `mpmath.iv`, 40 digits, outward rounding). Since Lemma 4.2 now proves it for every $q \geq 4$, the certificate is a redundant check of the proof and of the code; we record what it establishes.

Put $S_r(z) = \sum_{j \geq 0} j^r w_j z^j$ ($r = 0, \dots, 3$; $S_0 = L_q$) and $G(z) := S_0 S_1 - S_0 S_2 + S_1^2 = S_0^2 \kappa H_q / 4$ by (3.2). The certificate proves $G < 0$ on $(0, z_L^+]$ with $z_L^+ \geq z_L = \kappa_L^2$.

Small z , analytic. $G = \sum_{r \geq 2} g_r z^r$ with $g_2 = -(q^2 - q - 8)/((q+1)^2(q+2)^2(q+3)(q+4))$ (exact, `sym_reduction.py`) and, for $r \geq 3$, $|g_r| = |\sum_{i+j=r} w_i w_j j(1-j+i)| \leq r(r+1)^2((q+1)(q+2))^{-r} \leq (4/((q+1)(q+2)))^r$, using $w_i \leq ((q+1)(q+2))^{-i}$. With $y = z/((q+1)(q+2))$ and $c_2 = (q^2 - q - 8)/((q+3)(q+4))$,

$$G(z) \leq -c_2 y^2 + \frac{(4y)^3}{1-4y} < 0 \quad \text{for } 0 < y < y_0 := \frac{c_2}{64+4c_2}$$

(Lemma S; exact rational arithmetic). This settles $\kappa \rightarrow 0^+$, where interval enclosures of G would not resolve the sign. The continuation starts at $z_0(1-10^{-30}) < z_0 := y_0(q+1)(q+2)$, so there is no gap.

Enclosures. $S_r(z)$ is enclosed by the interval partial sum of N terms plus $[0, 2N^r w_N z^N]$, valid when $\rho_N = z/((q+2N+1)(q+2N+2)) \leq \frac{1}{16}$: the term ratios decrease, $(N+i)^r \leq N^r 8^i$ for $r \leq 3$, and $\sum_i (8\rho_N)^i \leq 2$. All series have positive terms, so this is a rigorous enclosure. z_L^+ is found by bisection until the interval lower bound of S_0 exceeds $q(q-1)/8$; since S_0 is increasing, $z_L \leq z_L^+$.

Continuation. $G' = (S_1^2 + S_0 S_2 + S_1 S_2 - S_0 S_3)/z$, so on $[a, b]$, $\sup |G'| \leq M(a, b) := (S_0 S_2 + S_1^2 + S_0 S_3 + 3S_1 S_2)(b)/a$ (all S_r positive and increasing), and $G \leq \bar{G}(a) + (b-a)M(a, b)$ on $[a, b]$ with $\bar{G}(a)$ the interval upper bound. Nodes are chosen adaptively (step $\approx |\bar{G}(a)|/(2M)$, halved until the bound is negative) and chain exactly from the first node to z_L^+ ; the bound certifies every real point of each $[a, b]$, not only the nodes.

Coverage and cost. Every q with $4 \leq q \leq 200$ is certified (`certify_4_100.log`, `certify_101_200.log`, all ok); $q = 4$ needs 52,820 nodes (68–86 s) because M is about 10^4 times larger than $\sup |G'|$ near z_0 (no cancellation in the majorant while $G' = O(z)$), larger q need 900–10,400 nodes and 6–131 s; the total is about 1.75 h. Earlier runs before a rounding fix of z_0 (`old_runs/`) give the same numbers digit for digit. The reviewer reran a byte-identical copy for $q = 4, 10, 37, 60$ (identical output), dumped the 52,820 nodes for $q = 4$ and checked that they chain ($\max |a_{i+1} - b_i| = 0$), recomputed at 80 digits with plain arithmetic the certified bounds at three nodes (e.g. node 0: $\bar{G} = -8.818 \cdot 10^{-8}$ against $G(a) = -8.818 \cdot 10^{-8}$, $M = 6.7 \cdot 10^{-2}$ against $\sup |G'| = 5.3 \cdot 10^{-6}$), Lemma S on 39 points of $(0, z_0)$ for $q = 4, 10$, and $S_0(z_L^+) - \frac{3}{2} = +1.4 \cdot 10^{-7}$ for $q = 4$.

Rigor caveats (from the review). Two comparisons — the final test $\bar{G}(a) + (b-a)M < 0$ and the ratio test $\rho_N \leq \frac{1}{16}$ — are evaluated in 40-digit floating point rather than in outward-rounded intervals; the slack in the first is at least $|\bar{G}(a)|/2 \geq 4 \cdot 10^{-8}$ against rounding of order 10^{-40} , and the second has slack by construction (N is increased by 2 beyond the first admissible value), so neither can change a verdict, but “every operation is outward-rounded” would overstate the code. The author’s script writes only per- q summaries; the reviewer’s patched copy (`certify_neg_steps.py`) writes the node list, included for $q = 4$.

7. Numerical table

Values from `num_phi.py` (mpmath, 60 digits, positive series; closed form and series agree to 10^{-20}), confirmed by the reviewer's independent series code to all printed digits. κ_Δ is the (numerically unique) sign change of Δ_q along the trajectory.

q	κ_L	κ_f	κ_Δ	$L_q(\kappa_f)$	$T(\kappa_f)$
4	3.469051	4.415714051	4.161910	1.9334	0.173
5	6.045574	7.613543394	7.290145	4.3697	1.415
6	8.174269	10.25463377	9.904743	8.478	3.448
10	15.384999	19.37353191	18.991063	62.88	14.02
20	30.744399	39.94164751	39.545885	2529	38.84
30	44.788910	59.99529121	59.597519	$6.8 \cdot 10^4$	59.86
40	58.215297	79.99966909	79.601279	$1.7 \cdot 10^6$	79.99
60	84.029603	119.9999987	119.601067	$9.6 \cdot 10^8$	120.00

The $q = 4$ fold $\kappa_f = 4.415714051$ is the value printed in 61Y0 Section 6 (whose normalizer $12(2 \cosh \kappa - 2 - \kappa^2)/\kappa^4$ equals L_4). Further observations, all numerical: $\kappa_f/q \rightarrow 2$ (to all printed digits from $q = 40$ on, consistent with $m_q \approx 1 - q/\kappa$ and $m_q(\kappa_f) \rightarrow \frac{1}{2}$: 0.3005, 0.3982, 0.4386, 0.4865, 0.4993, 0.5000 for $q = 4, 5, 6, 10, 20, 30$); κ_L/q rises from 0.87 ($q = 4$) to a maximum near 1.55 ($q \approx 12$) and decreases slowly (1.43 at $q = 50$, 1.26 at $q = 200$, 1.13 at $q = 1000$); the margin $H_q(\kappa_L)$ is $-4.2 \cdot 10^{-3}$ at $q = 4$ and grows in absolute value (-0.77 at $q = 1000$). The small $q = 4$ margin is why no estimate-based proof was found in [B3]; the argument of Section 4 does not see it. Sanity checks of every inequality used in the proof along trajectories (`verify_lyapunov.py`, 30 digits, $q \in \{4, 5, 6, 10, 30, 60\}$, 200-point grids up to $4q + 40$; the reviewer's `rev_num.py`, 50 digits, $q = 4, \dots, 60, 80, 100, 150, 200, 400, 300 + 200$ points): $Q < 0$ and $\frac{d}{d\tau}(e^{c\tau}Q) \leq 0$ on $(0, \kappa_L]$, with equality only in the limit $\kappa \rightarrow 0$ and at κ_L ; $Q > 0$ and $\frac{d}{d\tau}(e^{c\tau}Q) > 0$ on $[\kappa_f, 4q + 60]$; $T(\kappa_f) = \kappa_f^2 H'_q(\kappa_f)$ to 10^{-25} ; one sign change of H_q ; $\inf B = \frac{1}{4}$ approached as $\kappa \rightarrow 0$; the identity (3.1) agrees with a finite difference to 10^{-57} ; for $q = 2, 3$ with $c = 2, \frac{5}{2}$: $H_q > 0$, $Q > 0$, $\frac{d}{d\tau}(e^{c\tau}Q) > 0$ on $(0, 120]$. The Poisson identity (3.4) holds to 10^{-45} at three test points, and $|h - 4(\mathbb{E}J - \text{Var } J)| < 5 \cdot 10^{-17}$ relative on all grids.

8. Open questions

1. *Third-order normalizers in general.* The phase-lifted coherent orbits of [6XH6] and [6DJ3] have ${}_pF_{p+1}$ normalizers and Slater orbits have ${}_{k-1}F_k$ normalizers, all of order ≥ 3 . Does each admit an inhomogeneous lower-order equation playing the role of (E2), and is the forcing on the zero set of the fold numerator sign-definite with a threshold in a single monotone state variable, as in (3.1)? Not attempted.
2. *A unified criterion with the Riccati case.* For ${}_1F_1(a; c)$ the fold count is decided in the plane (m, κ) by transversality ([B3]: one fold iff $c > 2a$); here it is decided by an integrating factor in (X, u, κ) . Is there a single Lyapunov-type criterion, e.g. a condition on the Fano factor of the index law of a general power-series normalizer, that contains both?
3. *Grassmann rank $r \geq 2$* (matrix-argument ${}_1F_1$; 68BH open question 1): untouched.

4. Uniqueness of the sign change of T or Δ_q in (κ_L, κ_f) (Remark 4.4(ii)), and the exact leading coefficient of T on the zero locus (numerically about $\frac{3}{4}$ of $[\kappa^2]\Delta_q$): unimportant for the theorem, not derived.
5. A theorem on a single crossing of $\text{Var } J/\mathbb{E}J$ for power-series distributions with log-concave weight sequences, of which Proposition 3.4 with Theorem 4.1 is one instance.

9. Reproducibility and literature

All material is in `repro/` next to this file (Python 3.12, sympy 1.14, mpmath 1.3). `repro/author/` is a verbatim copy of the working directory `cycle3/work/B4_two_planes_fold/` (scripts, JSON results, the certificate logs `certify_4_100.log`, `certify_101_200.log`, the earlier runs in `old_runs/`, and `write_report.py`, which regenerates the working report). `repro/reviewer/` holds the adversarial reviewer's independent scripts and logs: `rev_sym.py` (symbolic- q re-derivation of (E2), (C1)–(C6), Lemma 3.3 for symbolic c , the derivative facts, (F1), g_2, g_3 , and $G = S_0^2 \kappa H_q / 4$ to $O(z^6)$), `rev_num.py` and `rev_fast.py` (independent series numerics), `rev_spot.py` (80-digit spot checks of the certificate), `certify_neg_copy.py` (byte-identical copy of the certificate) and `certify_neg_steps.py` (patched to dump nodes), with their logs and `rev_steps_q4.json`. The reviewer's files are copied unchanged except one marked line (REPRO) in `rev_spot.py`, where the saved copy left its node-index list empty; with the fix the rerun reproduces the three-node values quoted in Section 6. `repro/run_fast.sh` reruns the fast scripts on scratch copies and writes `repro/logs/*.log` and `repro/logs/timings.txt`; `repro/README.md` lists commands, runtimes and expected output.

script	certifies	arithmetic	time
<code>sym_reduction.py</code>	(E2) termwise, (E2)' = [B3] (8.1), (C1)–(C6), the bracket B , (F1), g_2 , g_3	sympy, symbolic q	5 s
<code>verify_lyapunov.py</code>	Section 7 sanity checks of every inequality of the proof	mpmath 30 digits	7 s
<code>num_phi.py q ...</code>	$\kappa_f, \kappa_\Delta, \kappa_L, u(\kappa_f), T(\kappa_f)$	mpmath 60 digits	16 s per q
<code>certify_neg.py q ...</code>	Section 6 certificate	mpmath.iv 40 digits	44–86 s ($q = 4$), 6–131 s otherwise
<code>margins.py,</code> <code>lemma_c_check.py,</code> <code>t_signchange.py</code>	exploratory numerics from before the proof (Fano margins; an estimate-based lemma covering $\kappa \leq 0.44q$; sign change of T)	mpmath	1, 1, 141 s
<code>reviewer/rev_sym.py,</code> <code>rev_num.py,</code> <code>rev_fast.py,</code> <code>rev_spot.py,</code> <code>certify_neg_copy.py</code>	the reviewer's independent checks listed above	sympy; mpmath 30–80 digits; mpmath.iv	177 s; 18 s (three q); 21 s; 3 s; 59 s ($q = 4$)

No step of the proofs depends on floating point: Theorem 4.1 and Corollary 5.1 use the identities of

Section 3, each a short hand computation machine-checked for safety, and Lemma 2.1.

Literature (web searches of 2026-09-14 by the author agent and the reviewer). Karp–Sitnik [KS] and Karp [Ka] prove log-concavity and Turán-type inequalities for Kummer and hypergeometric-like functions in their *parameters*; Baricz–Pogány [BP1,BP2] and Gaunt [Ga] prove monotonicity of ratios and functional inequalities for modified Struve and modified Lommel functions in the argument, via the Biernacki–Krzyż lemma on quotients of power series. None concerns the sign of $K' - \kappa K''$, the Fano factor of the index law, or a Lyapunov function of the present kind. Searches for over-dispersion of power-series distributions returned applied (neuroscience) hits only, and searches for the parity-conditioned Poisson law $\{K \geq q, K \equiv q\}$ returned only lower-truncated Poisson moment formulas; the identity (3.4) is elementary and appears to be new in this context. Generalized Curie–Weiss models with several transitions [EE] are the classical background of the radial problem; DLMF Chapters 11 and 13 [DLMF] are the standard references for the special functions named.

AI-assistance statement

The mathematics, code and text of this manuscript were produced by Claude Fable 5.1 agents (Anthropic) working under the direction of the author, who set the problem, selected the line of records and is responsible for the claims. One agent carried out the research (the reduction, the certificate, the proof, the scripts) and wrote the working report of 2026-09-14 on which this manuscript is based; a second, independent agent performed an adversarial review of that report with its own symbolic and numerical implementations, re-derived every step, reran and spot-checked the certificate, and found the corrections incorporated here (the bibliographic attributions of [BP2] and [Ga] and the range $q = 1, 2$ of the Struve case, the transcription $\kappa_L(40) = 58.215297$, the wording of Remarks 4.4(ii)–(iii), the rigor caveats of Section 6); a third agent wrote this text, assembled the reproducibility package and reran the fast scripts. No fallback to another model occurred in any of these sessions. The manuscript has not been peer reviewed by humans.

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